Design of Enterprise IoT

Tutorial 5 (Analyzing Security and Privacy Threats in IoT Systems)

The tutorial focuses on the understanding of various security and privacy threats that may occur to an IoT system in the sensing domain, fog domain, and cloud domain.

Concepts: Describe the concepts of Security and Privacy.

- What is Security and what is Privacy?
- How Security and Privacy differ from each other?
- Are there commonalities between Security and Privacy? If so, what are they?

Example: An IoT-based Smart Irrigation System (daCosta, 2013)

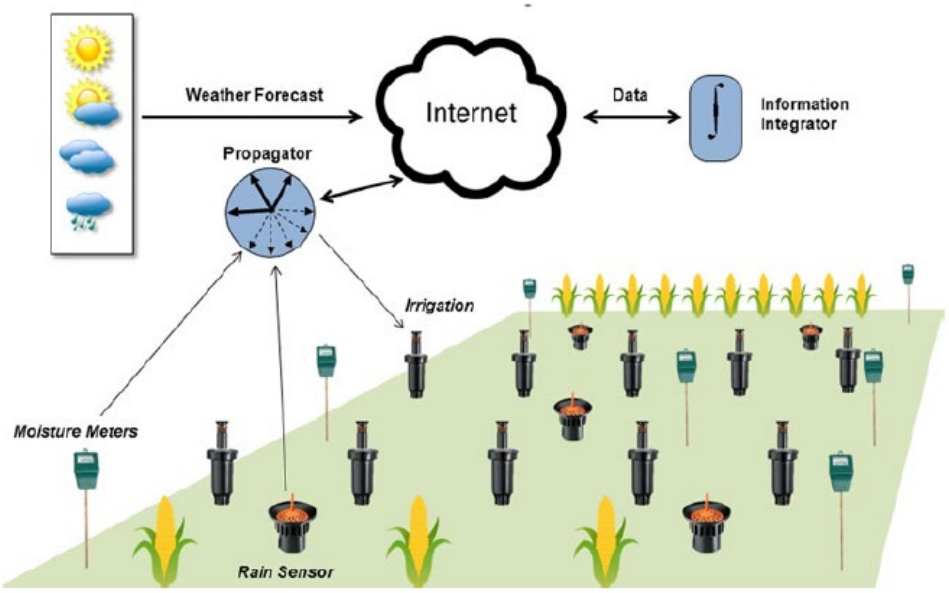


Figure 2-10. An integrator function retrieves data from end devices such as moisture sensors and external feeds such as the expected precipitation and humidity, using the information to control irrigation valves

An IoT-based smart irrigation system (as illustrated in the above figure) can consist of components and communication covering the following three domains:

IoT sensing domain: This domain is mainly composed of physical sensing devices, such as sensors and actuators, and transmits the data collected by these sensing devices to the devices in the fog domain (Rayes and Salam, 2022). The devices in the sensing domain include:

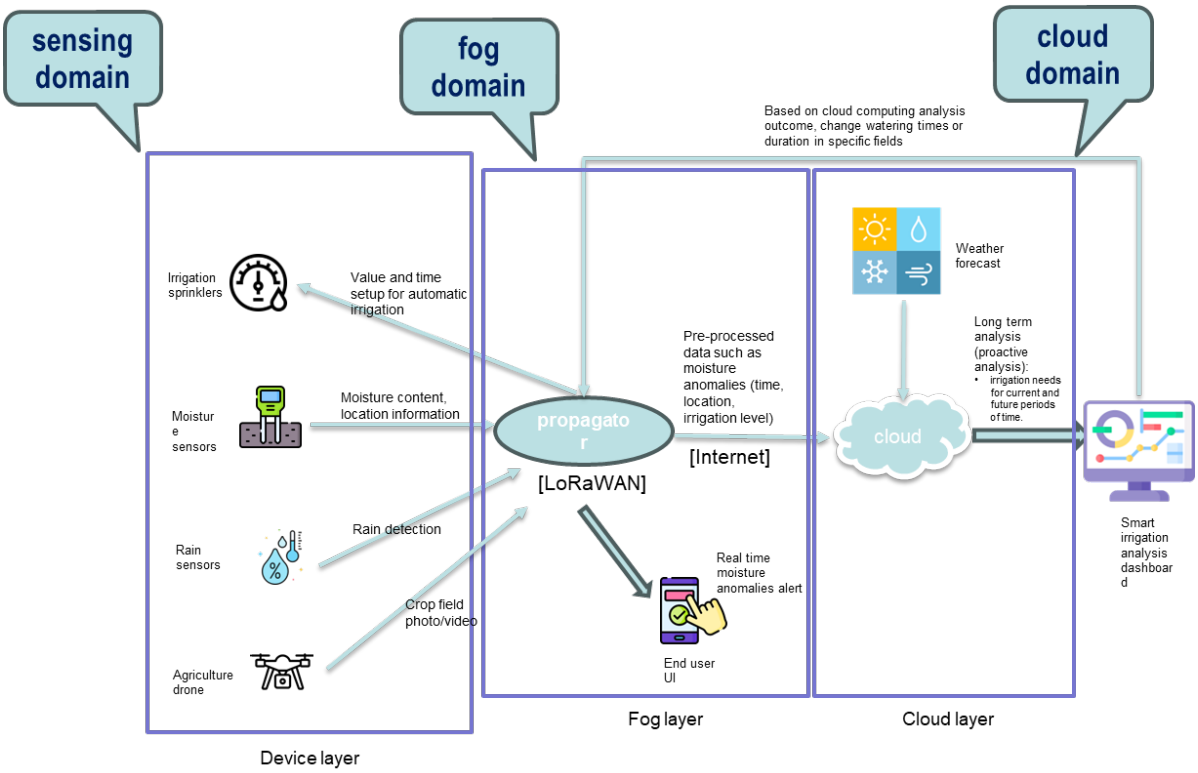
- Moisture sensor: measure soil moisture level, location of sensing area.
- Rain sensor: detect the rainfall conditions.
    - When it rains, the sensors are activated, which results in switching off the irrigation controller almost immediately.

- Irrigation sprinkler (actuator): activated to initiate irrigation.
- Agriculture drone:
  - Scan the crop field and collect a more global view of the terrain.
  - Check the irrigation status of the soil and monitor water needs (e.g., via infra-red technologies)

Fog domain: This domain consists of a set of fog devices, each performing operations such as aggregation, pre-processing and storage of data collected by the sensing devices. These fog devices are interconnected to facilitate communication among the sensing devices. Each fog device may also connect to one or more servers in the cloud domain.

Cloud domain: The domain consists of many high-capacity servers on which applications are deployed and responsible for performing complex computational analyses of data reported from fog devices.

Below is an example of an IoT connectivity diagram for the above solution:



While the IoT brings convenience, it also brings new security risks and privacy concerns, and these threats exist in each of these domains (Rayes and Salam, 2022). Security incidents may be accidental or intentional. They lead to security attacks in a broad sense. Animals, farm work, and machinery can easily access farming environments and cause incidents. In addition, IoT solutions comprise heterogeneous devices and software from distinct manufacturers installed between growth areas and the cloud. These specific features might cause several security vulnerabilities and could result in incidents that compromise IoT solutions (de Araujo Zanella et al., 2020).

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Activity: Please refer to the "*IoT security and privacy threat mapping template*" and list components and corresponding potential security and/or privacy threats in each domain and impacts on the IoT-based smart irrigation system described in the above use case example.

You can work on your document by answering the following question:

- What threats could happen in each domain (including sensing, fog, and cloud domain)?
- What are these threats (components and examples)?
- What security requirements would be violated if such threats happen?
- What are the impacts of each attack on your IoT solution?

[Hint]

You may consider the following key items when answering questions required in the threat mapping template.

- Sensing domain
    - What are the functionalities of each of the devices?
    - What information does each of the devices have access to? (e.g., whether it includes sensitive, personal information)
    - What are the characteristics of each of the devices (e.g., it is useful to consider from five perspectives of IoT solution: constrained, unattended, diversity, interdependence, and ubiquitous)?
    - What are the components that support network connection of these devices?
- Fog domain
    - What types of fog domain security/privacy threats are relevant?
        - What fog devices are included?
        - How do sensing devices and fog devices interact with each other?
        - What communication protocols are in use?
    - What security/privacy attacks are possible in fog layer applications (if applicable)?
- Cloud domain
    - What types of cloud domain security/privacy threats are relevant?
    - What security/privacy threats are possible in the cloud layer application (if applicable)?

References:

- Rayes, A. & Samer, S. (2022). Internet of Things from Hype to Reality: The Road to Digitization (3rd eds). Spinger.
- daCosta, F. & Henderson, B. (2013). Rethinking the Internet of Things: A Scalable Approach to Connecting Everything. Apress.
- de Araujo Zanella, A. R., da Silva, E., & Albini, L. C. P. (2020). Security challenges to smart agriculture: Current state, key issues, and future directions. Array, 8, 100048.